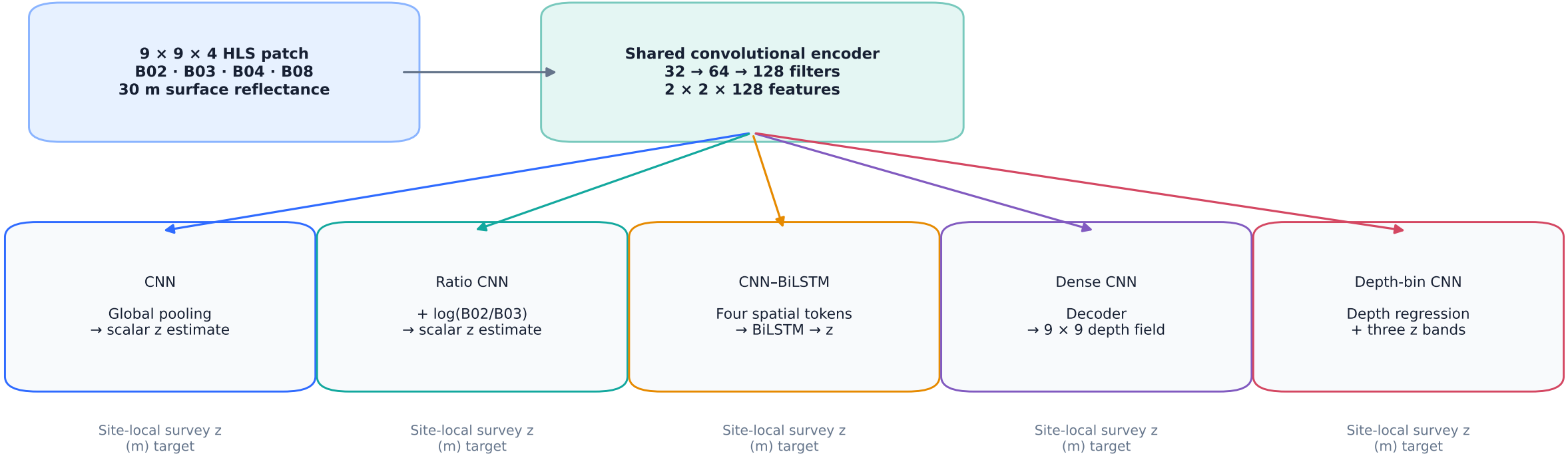
Model families and common HLS inputs



Spatial evaluation: three buffered alongshore zones · one label per 30 m pixel
Baselines: Stumpf log-ratio regression and random forest · held-out pixels are never used for fitting